

Neural Pool: Mini Math Document

Project Pipeline (Data $\rightarrow$ VAE $\rightarrow$ DiT + Diffusion Forcing)

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1 Scope and Current Configuration

This document summarizes the math used in the current codebase and active training setup.

Item	Value (current)
Raw frame size	$H \times W \times 3 = 72 \times 128 \times 3$
Action size	$A = 3$ (force_x, force_y, trigger)
Latent size per frame	$C \times H_z \times W_z = 4 \times 9 \times 16$
Latent scalars per frame	$d_z = 4 \cdot 9 \cdot 16 = 576$
Compression (element ratio)	$(72 \cdot 128 \cdot 3)/576 = 48\times$
DiT context length	$L = 8$
DF rollout steps	$K = 4$
Diffusion steps	$T = 1000$
DiT width/depth	$d_{\text{model}} = 2048$, heads = 32, layers = 30
DiT parameter count	1,520,566,848

2 Data Tensors

For episode e and frame index t :

$$x_{e,t} \in \{0, \dots, 255\}^{72 \times 128 \times 3}, \quad (1)$$

$$a_{e,t} \in \mathbb{R}^3, \quad (2)$$

$$s_{e,t} \in \mathbb{R}^{16 \times 4}. \quad (3)$$

Action vector:

$$a_{e,t} = [f_x, f_y, \text{trigger}], \quad (4)$$

where trigger is 1.0 on shot-release frames, otherwise 0.0.

Simulation state per ball stores:

$$[\text{pos}_x, \text{pos}_y, \text{vel}_x, \text{vel}_y]. \quad (5)$$

3 VAE

3.1 Encoder/Decoder Geometry

The encoder downsamples spatially by factor 8 (three stride-2 stages):

$$72 \times 128 \rightarrow 36 \times 64 \rightarrow 18 \times 32 \rightarrow 9 \times 16. \quad (6)$$

Approximate posterior:

$$q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x))), \quad \sigma_\phi(x) = \exp(\tfrac{1}{2} \log \sigma_\phi^2(x)), \quad (7)$$

with reparameterization:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (8)$$

Decoder:

$$\hat{x} = g_\theta(z), \quad z \in \mathbb{R}^{4 \times 9 \times 16}. \quad (9)$$

3.2 VAE Loss

The implemented loss is:

$$\mathcal{L}_{\text{VAE}} = \lambda_{L1} \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \mathcal{L}_{\text{LPIPS}}(x, \hat{x}) + \lambda_{\text{KL}} \beta(n) \mathcal{L}_{\text{KL}}, \quad (10)$$

where

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \mathbb{E} [1 + \log \sigma^2 - \mu^2 - \sigma^2], \quad (11)$$

$$\beta(n) = \beta_{\text{start}} + \min\left(1, \frac{n}{N_{\text{warmup}}}\right) (\beta_{\text{end}} - \beta_{\text{start}}). \quad (12)$$

Current values:

$$\lambda_{L1} = 1.0, \lambda_{\text{LPIPS}} = 0.1, \lambda_{\text{KL}} = 1.0, \beta_{\text{start}} = 10^{-6}, \beta_{\text{end}} = 5 \times 10^{-4}, N_{\text{warmup}} = 300,000 \text{ frames}. \quad (13)$$

4 Latent Training Samples

After VAE export, latent shards provide:

$$z_{e,t} \in \mathbb{R}^{4 \times 9 \times 16}, \quad a_{e,t} \in \mathbb{R}^3. \quad (14)$$

Baseline training sample at time t :

$$\text{context}_t = [z_{t-L+1}, \dots, z_t] \in \mathbb{R}^{L \times C \times H_z \times W_z}, \quad (15)$$

$$\text{action}_t = a_t \in \mathbb{R}^3, \quad (16)$$

$$\text{target}_t = z_{t+1} \in \mathbb{R}^{C \times H_z \times W_z}. \quad (17)$$

Diffusion-forcing sample additionally uses:

$$\text{actions_seq}_t = [a_t, \dots, a_{t+K-1}], \quad \text{targets_seq}_t = [z_{t+1}, \dots, z_{t+K}]. \quad (18)$$

5 Diffusion in Latent Space

Define linear noise schedule:

$$\beta_1, \dots, \beta_T \text{ linearly spaced from } 10^{-4} \text{ to } 0.02, \quad \alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i. \quad (19)$$

Forward corruption of clean latent x_0 :

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (20)$$

Model predicts $\hat{\epsilon}_\theta$ and baseline objective is:

$$\mathcal{L}_\epsilon = \mathbb{E}_{t, \epsilon} [\|\hat{\epsilon}_\theta - \epsilon\|_2^2]. \quad (21)$$

Estimated clean latent from $(x_t, \hat{\epsilon})$:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}. \quad (22)$$

6 Action-Conditioned DiT

Flatten each latent frame:

$$\text{vec}(z) \in \mathbb{R}^{d_z}, \quad d_z = 576. \quad (23)$$

Tokenization:

$$h_i = W_c \text{vec}(z_{t-L+i}) \in \mathbb{R}^{d_{\text{model}}}, \quad i = 1, \dots, L, \quad (24)$$

$$h_{\text{tgt}} = W_t \text{vec}(x_t) + E_a(a_t) + E_t(t) \in \mathbb{R}^{d_{\text{model}}}. \quad (25)$$

Transformer input:

$$H = [h_1, \dots, h_L, h_{\text{tgt}}] + P, \quad (26)$$

then

$$\hat{\epsilon} = \text{unvec}(W_o \text{Transformer}(H)_{\text{last}}) \in \mathbb{R}^{4 \times 9 \times 16}. \quad (27)$$

7 Diffusion Forcing (DF)

With rollout length K , for each rollout step $k = 1, \dots, K$:

$$\mathcal{L}_k = \|\hat{\epsilon}_k - \epsilon_k\|_2^2. \quad (28)$$

Teacher forcing probability as a function of processed samples n :

$$p_{\text{tf}}(n) = p_0 + \min\left(1, \frac{n}{N_d}\right) (p_1 - p_0), \quad (29)$$

with current values:

$$p_0 = 0.98, \quad p_1 = 0.05, \quad N_d = 420,000,000. \quad (30)$$

Context update between rollout steps:

$$\tilde{z}_{k+1} = m_k \odot z_{k+1}^{\text{gt}} + (1 - m_k) \odot \hat{x}_{0,k}, \quad m_k \sim \text{Bernoulli}(p_{\text{tf}}). \quad (31)$$

Then

$$\text{context} \leftarrow [\text{context}_{2:L}, \tilde{z}_{k+1}]. \quad (32)$$

Final DF objective:

$$\mathcal{L}_{\text{DF}} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_k. \quad (33)$$

Implementation detail currently enabled:

$$\hat{x}_{0,k} \text{ is detached before context insertion.} \quad (34)$$

8 Distributed Batch Math

If per-rank batch is B_{local} and world size is R :

$$B_{\text{global}} = B_{\text{local}} \cdot R. \quad (35)$$

Current run uses $B_{\text{local}} = 512$, $R = 2$, so $B_{\text{global}} = 1024$.

9 What We Are Using

The current world-model training mode in this repository is:

$$\text{DiT} + \text{Diffusion Forcing} \quad (36)$$

with latent-space diffusion loss, action conditioning, rollout training ($K = 4$), and decayed teacher forcing.